# FRASS-FVS INTEGRATION SPRINT 1

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# INTRODUCTION

- Team: Ziang Chen, Meshach Erisca, Tomas Stoker
- Client: Dr. Bill Schlosser, D&D Larix
- Goal: Integrate FVS into the FRASS workflow

# SPRINT OBJECTIVES

- Gain an understanding of the FRASS system and FVS
- Document requirements
- Set up dev environments
- Learn R and FVS
- Set up a first run

# REQUIREMENTS GATHERING PROCESS

- Weekly Meetings with Client, have had four meetings.

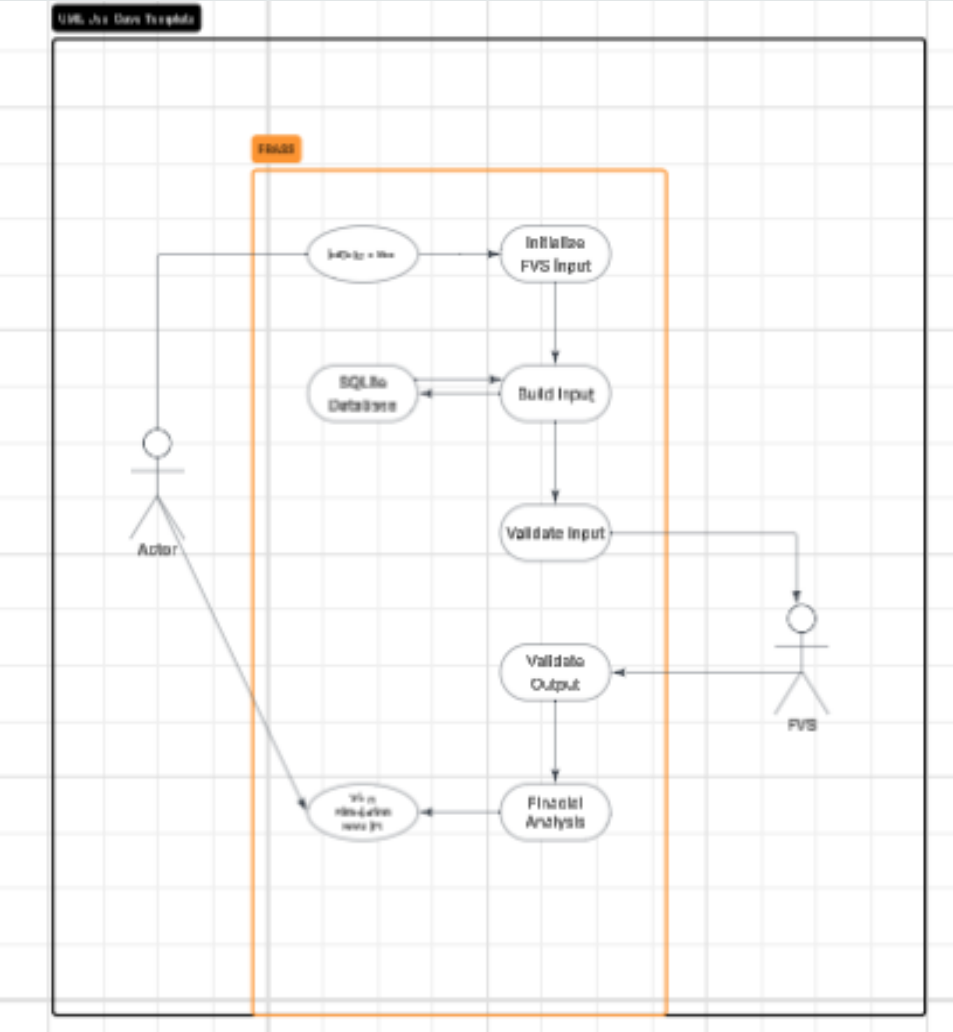
Meetings included history and baseline need for the system, Contribution overview from previous teams and Consultants, as well as several brainstorming sessions on implementation requirements.

# DOCUMENTATION OF USER STORIES

User Stories:

- **Initiate a Base Run**
  - **Gather Information**
  - **Validate Required Fields**
  - **Format Data**
  - **Pass Data to FVS**
- **Review a Projection**
  - **Given FVS Output**
  - **Search for Initial and Simulated Data Points that Produced Run**

# DOCUMENTATION OF USE CASE DIAGRAM




| | |
|---|---|
| *Actors:* | User |
| *Pre-condition:* | Stand data within FRASS database |
| *Post Condition:* | Simulation Report is generated based on user parameters |
| *Path:* | 1.Initiate Run<br>2.Initialize Input Table<br>3.Build Input from Internal Database<br>4.Validate Input<br>5.Start External Run<br>6.Validate Output<br>7.Save Results<br>Handoff Control to Main System |

# INSTALLATION AND EVAL OF EXISTING SOLUTION

- The client is currently expecting users to bring their own FVS data.
  - This clashes with FRASS data collection like the Flex Cruiser.
  - The users are forced to leave the FRASS ecosystem.
- There was a previous attempt to integrate FVS with FRASS using Fortran code, but it was abandoned.
  - FVS Fortran code is not directly compatible with FRASS C# code.

# KANBAN OVERVIEW AND CONTRIBUTIONS

# FEATURE DEMONSTRATION

- Completed Features: Calling FVS and R-FVS using R Commands

# SKILLS IDENTIFICATION AND LEARNING

- Ziang
  - Learned existing FRASS workflow
  - Communication and organization
  - FVS
- Meshach
- Learned general use and features of FRASS
- Learned structure of FVS and FVS Online
- Basic understanding of R commands
- Tomas
  - Learned General Overview of FRASS Resource
  - Learned General Overview of FVS and Use
  - Initial Tutorials for Learning R

# RESPONSIBLE USE OF AI

- Application
  - Used to quickly search up info
  - Rephrase sentences to be more professional
  - R tutorial
- Limitations
  - Checked all outputs

At the end of the day, the team takes ownership for final decisions and deliverables

# NEXT STEPS AND SPRINT RETROSPECTIVE

- What went well
  - Understand the project from meetings
  - Documented requirements
- What we'll improve
  - Time management
  - More proactive on documentation
  - Meetings

# CONCLUSION

- Sprint 1 established requirements and an understanding of the system